

Cybersecurity (in Data Science)

2026 – Lecture 10 – Anomaly-Based Detection

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Course Plan



Week	MONDAY	Content (10:45-12:30)	Obs	TUESDAY	Content (10:45-12:30)
1	31-Aug	Intro + Basics of Cybersecurity		1-Sep	Lab 1 - Setup and Questions
2	7-Sep	Malware & Attack Frameworks		8-Sep	Lab 2 - Attack Analysis
3	14-Sep	Networks & Forensics		15-Sep	Lab 3 - Wireshark Forensics
4	21-Sep	Web Security		22-Sep	Lab 4 - WebGOAT XSS/SQLi
5	28-Sep	Cryptography, Hashing		29-Sep	Lab 5 - Digital Signatures
6	5-Oct	Authentication, Access Control	Filippo Scaramuzza →	6-Oct	Lab 6 - Authentication & AC
7	12-Oct	Data Protection & Data Privacy	Jessica de Pascale →	**13-Oct**	Lab 7 - Protection & Privacy
8	19-Oct	Recap Session (1)		20-Oct	Recap Session (2)
	26-Oct	**Exam (09:00 – 12:00)**	Daniel Grimm →	**27-Oct**	Intrusion Detection & Signat.-Based
	2-Nov	Anomaly-Based Detection	- Deadline Groups -	3-Nov	Anomaly-Based Detection
11	9-Nov	Project Feedback	Time changed	**10-Nov**	Project Feedback **(14:00 – 18:00)**
12	16-Nov	Project Feedback		17-Nov	Guest Lecture – Martijn Sprengers
13	23-Nov	Project Feedback	maybe shorter →	24-Nov	Project Feedback
14	30-Nov	Project Presentation**		1-Dec	Project Presentation**
	18-Dec	**Resit (13:30 – 16:30)**			

Red/Underscored: Different Lecturer

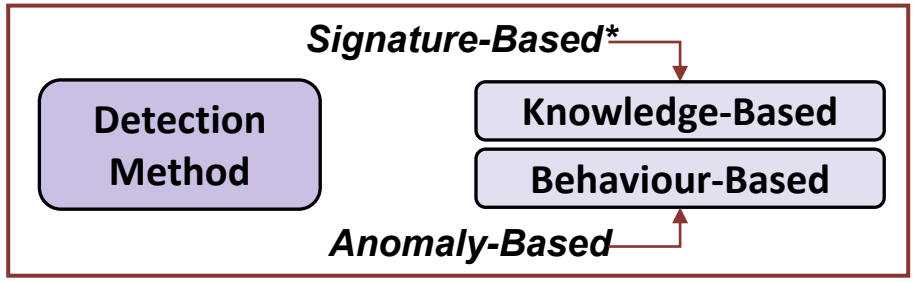
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** To Change Based on Number of Groups

- Content
- Lab
- Project
- Exam
- Recap
- Guest Lecture

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Previous Lesson: IDS Taxonomy



Detection Method – How to detect Intrusions
- Knowledge: Information about Attacks
- Behaviour: Normal behaviour of system

IDS

Knowledge-Based: Information about Attacks
- Defines how an attack happens (Pattern, Signature, etc)
- Then, tries to match with incoming audits

Types
- Signature-Based: Describes “exactly” how it happens
- Modelling: Maps similar flows
- Machine Learning: Learns the pattern

Previous Lesson: Knowledge-Based

Good if

1. You know the attack
2. You know its variations
3. You have enough data to test it

But what about the attacks you do not know? (they are the majority)

- Zero-day
- Small deviations
- etc

Knowledge-Based: What you know

“

Knowing yourself is the beginning of all wisdom;

-- Aristotle

”

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Previous Lesson: Knowledge-Based

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3. You have enough data to test it

But what about the attacks you do not know? (they are the majority)

- Zero-day
- Small deviations
- etc

Knowledge-Based: What you know

Anomaly-Based: What you don't know

“

True wisdom comes from
understanding one's own ignorance;

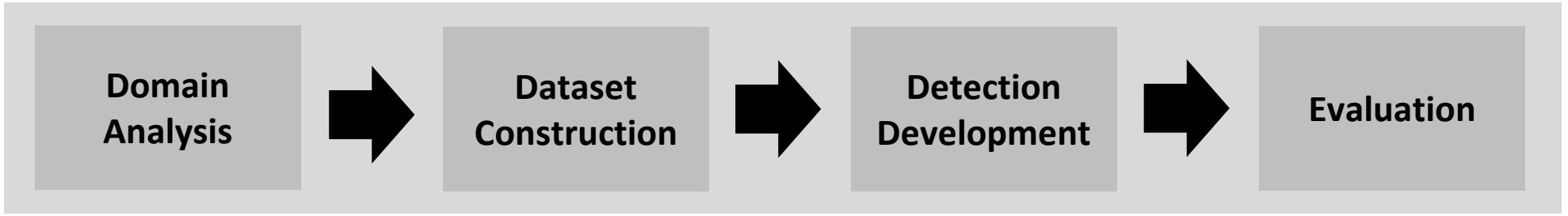
-- Plato

”

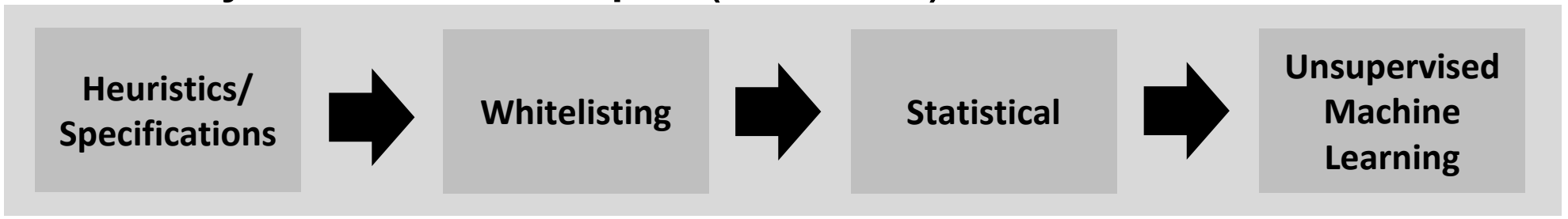
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Agenda

1. Anomaly-based Intrusion Detection Engineering (Today)



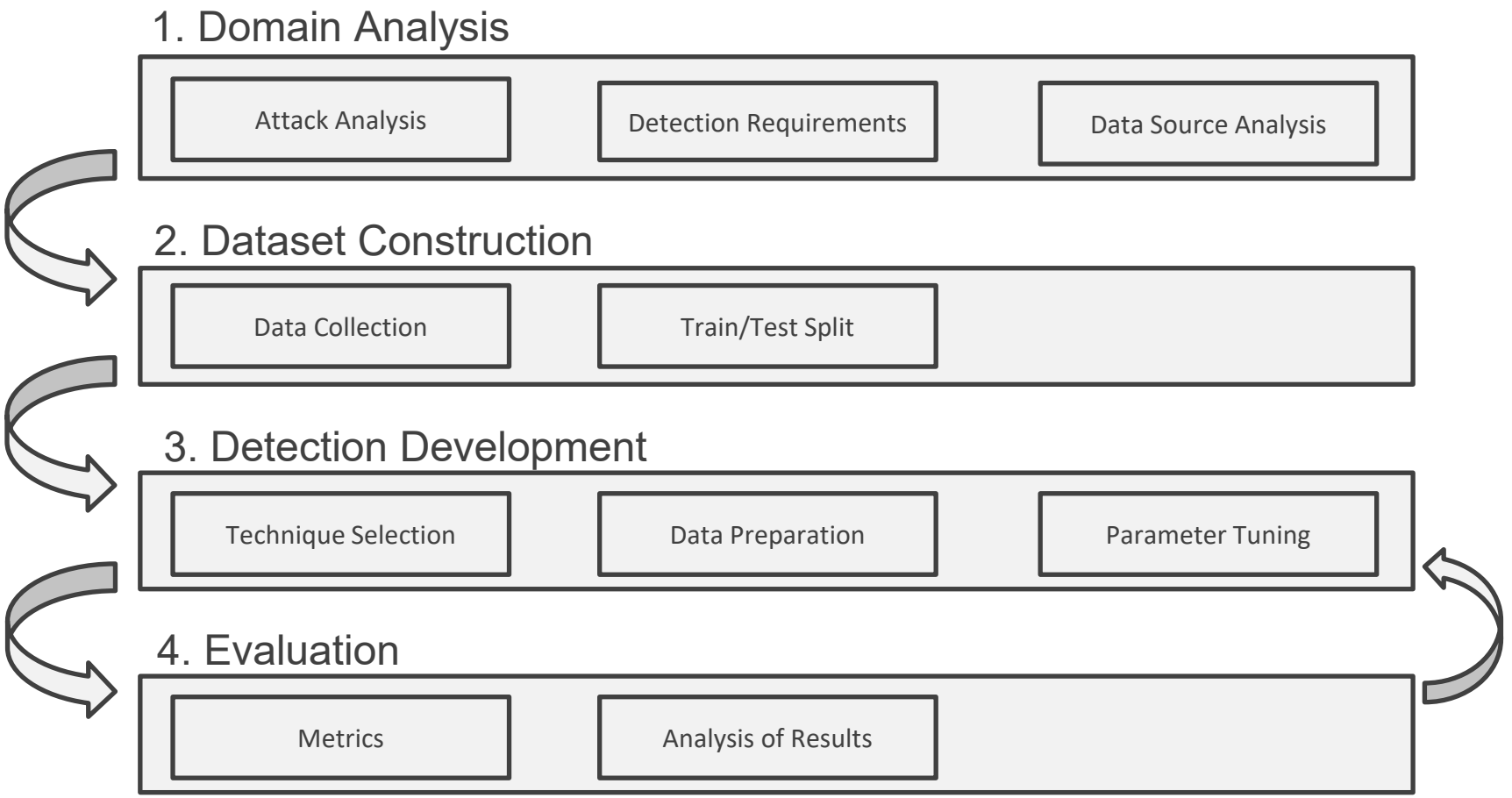
2. Anomaly Detection Techniques (Tomorrow)



Anomaly Detection

Anomaly Detection Engineering

Anomaly Detection Engineering Workflow



DISCLAIMER

Anomaly Detection requires creativity.

Guidelines in the slides with this icon should help you approach the Project:



Anomaly Detection

Domain Analysis

Domain Analysis – Attack Technique Analysis



Define **tractable** but **general** scope

- Example: Detect SQL injection (SQLi)

  → Tractable: SQL injection in HTTP data.

  → General: SQLi achievable in multiple ways, more than signatures can capture.

How SQL Injection happens in general?

- Read CTI reports to understand it (see Lesson 2)

Identify variations:

- Check tools or modus operandi.

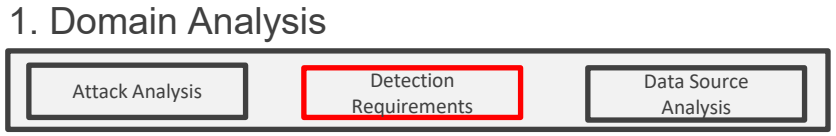
Organise variations (attack vectors, procedures)

Identify commonalities across variations using the classification.

Project

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Domain Analysis – Detection Requirements



Functional Requirements

- Acceptable False Alerts / Missed Attack rates
- Granularity of Detection & Report

  -- Every instance (like SQLi) or a summary (like DDoS)
- Target/Attacker Information Necessary for Response

  -- Precise (IP, username, etc) or Generic (analyst can figure it out)

Non-Functional Requirements

- Performance Constraints (based on data volume)
- Other Constraints (real-time or batch)

Reference

Domain Analysis – Data Source Analysis



Identify where attack behaviour can be observed in the data

- Review sources
- Scope-down data

} See previous lesson: Signatures

Formulate Hypotheses: How attack and normal data may differ

Identify features that allow testing hypotheses:

- Dimensions (e.g., host, user, command, time of day)
- Metrics (e.g., count, min/max, entropy)?
  → Is a single metric sufficient?
  → Correlate multiple ones (e.g., detect count of X decreases when sum of Y increases)?

Note: Features should be selected for a reason
"The more the merrier" usually does not work (more noise than signal)



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Anomaly Detection

Dataset Construction

Dataset Construction – Data Collection



Relevant audit data (PCAPs/logs):
→ includes benign and malicious behaviour

Realistic **Benign** (normal) data:
- Collection time must account for data fluctuations over time
- Simulated environments miss intricacies of reality
  → Oversimplified strategies.
  → Overestimated accuracy

e.g., benign outliers
- usage fluctuations on real user behaviour
- vs. simulated flat usage

Reference

Dataset Construction – Data Collection



Relevant audit data (PCAPs/logs):
→ includes benign and malicious behaviour

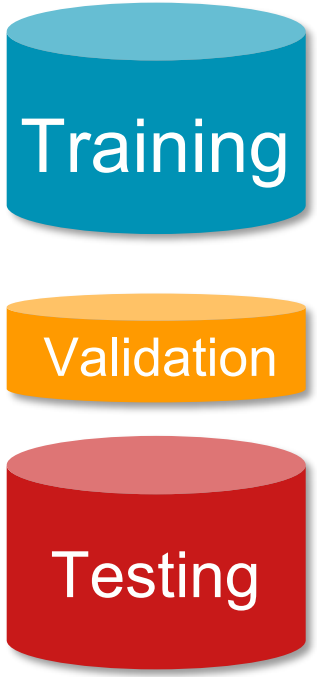
Realistic **Malicious** data (e.g., from traces of real attacks):

- Mix malicious / benign
  → Unnatural ratio can mislead detection
  → Attacks rare in reality? Make them rare in the dataset
- Careful injection
  → Leftover artefacts may change (apparent) difficulty
- Balanced attack data sampling
  → Instance with way more samples than another → Misleading evaluation

In practice: No control over data collection (Still important to understand the process)

Reference

Dataset Construction – Train/Test Split



Training: Explore/build detection model

- Sufficient for learning to converge (e.g., check loss function)
- No need for 100% attack free
- Must temporally precede testing data
  Otherwise, model "knows the future" (unrealistic assumption)



Validation [optional]: **Tune** detection parameters (e.g., thresholds).

- Allows checking converging without overfitting model to training data.

Testing: Evaluate ability of model to identify attacks.

- Should include data not in training/validation
  E.g., evolution of normal data, attack variations

Prepare dataset

1. Explore data to confirm/refine the hypotheses
2. Label benign and attack samples:
   -- may require manual or semi-automated work (e.g., signatures).



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Anomaly Detection

Detection Development

Technique Development - Selection



General Guidelines

1) Choosing technique (family):

- One that adapts to hypothesis' context
- Has underlying assumptions compatible with data
  (e.g., underlying data distribution, clean training dataset, etc.).

2) If unsure, test different techniques with small representative dataset
→ Evaluate which fits assumptions best

3) Prefer Techniques:

- Simpler over complex ones (if they do the job)
- That provide explainable results.

4) If needed, combine different techniques

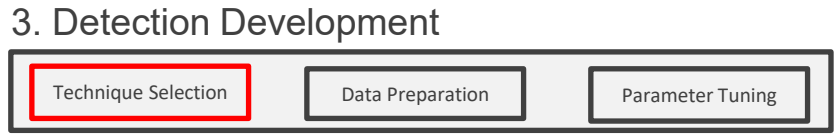


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Technique Development - Selection

Basic Technique Family Selection Guideline

Is it possible/practical to	Techniques Based on
Manually specify how data should be checked for normality?	Heuristics/Specification
Detected anomalies as a missing element in 1+ sets of previously learnt variable values?	Whitelisting/Allowlisting
Detected the anomaly as outliers in one (or more) uncorrelated variables?	Statistical
Detected the anomaly as outliers in multivariate datasets?	Machine Learning



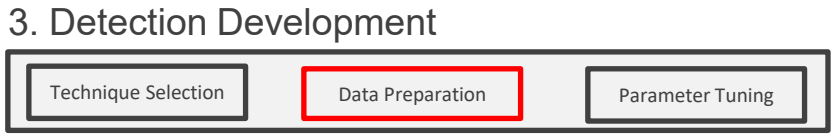
More about the techniques on the Second part of this content

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Technique Develop. – Data Preparation

Modify the data to a format usable with (ML) Technique selected.



1) Feature Encoding

Categorical Features (discrete): Often require encoding

- Ordinal (natural order): Ordinal encoding — Examples: high = 3, medium = 2 , low =1
- Nominal: One-hot encode (binary column) — TCP = [1,0,0], UDP = [0,1,0], HTTP = [0,0,1]
  Target encoding (if high cardinality) — TCP = 0.2, UDP=0.5, HTTP=0 (% malicious)

Numerical Features: *May* require scaling

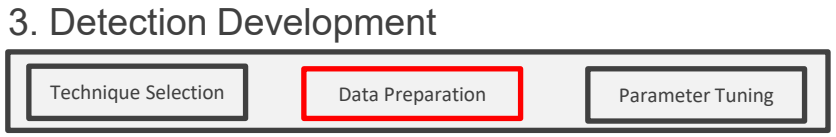
- Interval: In case there is no true zero value — Temperature in Celsius (0 is not 'no temperature)'
  Helps normalise the range
- Ratio: has a zero value. — Weight, income
  Often applied for consistency



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Technique Develop. – Data Preparation

Modify the data to a format usable with (ML) Technique selected.



2) Feature Scaling

- Adjusting range/distribution for comparable magnitude
- Prevents one feature to "weight" more than others because of its scale.

KNN, SVM, or gradient descent-based models: sensitive to differences in scale

3) Feature Derivation

Allows leveraging "hidden" values for learning/detection.

- Calculate metrics over rolling time window
- Add domain knowledge to features (e.g., IP reputation, group semantically-similar values)
- Calculate entropy of a string or payload
  → measures randomness of data

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Technique Develop. – Parameter Tuning

Find optimal values for configuration parameters of the technique



Parameter Tuning

- Often: configure at least a threshold
- Defined empirically using validation (or training) dataset
- Typical techniques for tuning:
  1) Manual. Good if you:
     Understand the model well
     Have limited parameters to test
  2) Grid (set of all possible combinations).
     Test and pick one
  3) Random Search (randomly sample values)
     Surprisingly effective — often finds a near-optimal configuration faster

Model	Common Parameters to Tune
Local Outlier Factor (LOF)	n_neighbors, contamination, metric, leaf_size
Isolation Forest (IForest)	n_estimators, max_samples, contamination, max_features
One-Class SVM (OCSVM)	kernel, nu, gamma



Anomaly Detection

Evaluation

Evaluation – Metrics

Confusion/Error Matrix: Table visualization of predicted vs. actual (ground truth) labels.

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Precision = TPs / (TPs + FPs)
→ How precisely it identifies attacks (quality).

Recall/TPR (Coverage) = TPs / (TPs + FNs)
→ How many attacks are correctly identified

F1-score: 2 * Precision * Recall / (Precision + Recall)
Harmonic mean of Precision and Recall.

Accuracy = (TPs + TNs) / (TPs + FPs + TNs + FNs)
Number of correct predictions divided by total number of predictions.

FPR = FPs / (FPs + TNs)
Measures how many benign samples are incorrectly identified by the IDS.

TNR (Specificity) = TNs / (FPs + TNs)
Measures how many benign samples are correctly identified.



Evaluation – Analysis of Results

Investigate FNs and FPs

- Understand their occurrence
- Tweaking detection parameters is not the only solution

You may discover that:

- Assumptions (from hypotheses) do not hold
- Chosen technique is not sufficiently robust

Is this good?

- False Positive Rate (FPR) = 0.01
- Precision = 0.90

Interpretation depends on how dataset was built

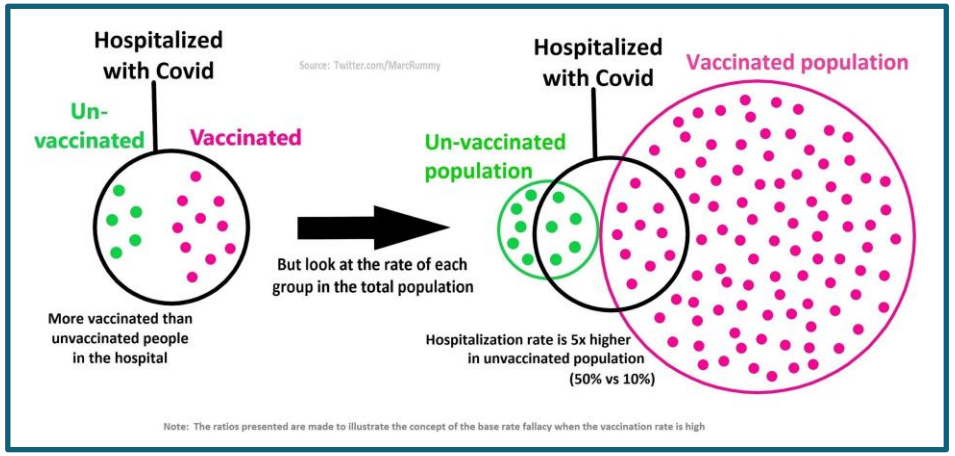


Evaluation – Analysis of Results

These may seem good…

1) FPR = 0.01

- Assume 1M benign samples/day
- This means 10K false alerts a day
  → Alert fatigue



Base-rate Fallacy

2) Precision = 0.9

- Good if dataset has a 50/50 split (benign and malicious) samples
- Reality: daily ratio of 1/1M
  → Actual Precision is bad as 2e-7.

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Next

Tomorrow (03-Nov-25)

- Anomaly Detection Lesson 2 (Techniques)

Following Weeks

- Week 11: Project Feedback
- Week 12: Project Feedback
- Week 13: Project Feedback & Guest Lecture (25/11)
- Week 14: Project Presentations